SQL Server Edition Optimisation – Architecture & Approach

This document outlines a scalable, secure approach for identifying SQL Server Enterprise to Standard downgrade opportunities using Azure Arc and Azure-native services.

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1. OBJECTIVE

Enable large-scale analysis of SQL Server estates to determine whether Enterprise Edition is required, and identify downgrade opportunities without impacting performance.

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2. KEY PRINCIPLES

- Feature usage is the only hard blocker to downgrading

- Performance constraints can be solved through infrastructure scaling

- Analysis must be data-driven and automated

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3. DATA SOURCES

A. Azure Resource Graph

- Provides instance inventory
- Captures edition, version, cores
- Enables estate-wide discovery

B. DMV / T-SQL Extraction

- Identifies Enterprise feature usage
- Uses sys.dm_db_persisted_sku_features
- Must be executed inside the SQL engine

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4. EXECUTION MODEL (NO AMA REQUIRED)

Recommended approach:

- Use Azure Arc Run Command to execute DMV scripts locally
- Avoids direct SQL connectivity
- No additional agents required

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5. DATA INGESTION OPTIONS

Primary:

- Logs Ingestion API
- Secure API-based push to Log Analytics

- Uses Entra ID authentication and RBAC

Alternative:

- Azure Blob Storage (for lightweight ingestion pattern)

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6. SECURITY MODEL

- No SQL credentials distributed
- Scripts executed locally via Arc agent
- HTTPS outbound connectivity only
- RBAC-controlled ingestion via API

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7. DATA PIPELINE

1. Query Resource Graph for SQL instances
2. Execute DMV scripts via Arc Run Command
3. Collect results centrally
4. Push structured JSON via Logs Ingestion API
5. Store in Log Analytics custom table
6. Model using KQL
7. Visualise in Power BI

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8. POWER BI OUTPUT

Dashboard includes:

- Instances by edition
- Feature usage classification
- Downgrade eligibility
- Estimated cost savings

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9. FINAL ARCHITECTURE

Arc-enabled SQL Servers

-> Run Command executes DMV queries

-> Central orchestrator collects results

-> Logs Ingestion API pushes data

-> Log Analytics stores results

-> Power BI visualises insights

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10. KEY TAKEAWAY

Azure Resource Graph cannot identify Enterprise feature usage.

A combination of:

- ARG (inventory)

- DMV extraction (feature detection)

- API ingestion (data pipeline)

is required to build a scalable optimisation solution.